# XAVIER DELFOSSE

**Data Scientist & Life Science Engineer**


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## EXPERIENCE

### Independent Contractor: Machine Learning and Data Science Specialist & Lead Data team

**Tradepedia**

May 2024 – July 2025 Remotely

- Designed and developed a reinforcement learning-based model for stock market analysis and prediction.
- Deployed MLflow for model tracking and versioning; served models via custom FastAPI endpoints.
- Mentored junior data scientists and managed team collaboration via GitHub.
- Assumed project management responsibilities, including backlog and roadmap planning.

### Bioprocess Development Scientist

**Securecell AG**

Oct 2021 – August 2024 Urdorf, ZH

- Developed and tested automated control strategies for mammalian and microbial cell cultures using real-time process data.
- Integrated convolutional neural networks and machine learning models into bioprocess pipelines.
- Authored and executed validation test plans for new bioprocessing products.
- Led various bioprocess experiments (cell culture and fermentation) and contributed to the Bioprocess 4.0 product roadmap.

## THESIS PROJECTS

### Master Thesis

- "Implementation and usage of an electronic nose for the monitoring of bioprocesses," 2024.

### Bachelor Thesis

- "Characterization, optimization, and validation of an autosampling unit for microbial fermentation processes," 2021.

## LANGUAGES

**French**
**English**
**German**

## EDUCATION

### Ph.D. in Life Sciences

**HES-SO Valais Wallis / Karlsruhe Institute of Technology**

2026 – 2029 Sion

### M.S. in Applied Computational Life Sciences

**Zurich University of Applied Sciences**

2021 – August 2024 Wädenswil

### B.S. in Life Science Technologies option Biotechnology

**HES-SO**

2018 – 2021 Sion

## HOBBIES

**Sports**
Bouldering, Biking, Roller Skating and Calisthenics

**Games**
Competitive and certified judge for Pokémon TCG (2009–2022)

**Others**
Homebrewer, hobbyist with Arduino and Raspberry, 3D printing

## SKILLS

Python | R | SQL | Matlab | PyTorch | TensorFlow | Scikit-learn | ML | DL | RL | NLP | LLMs | LangChain | Langflow | MARL | MLflow | Git

MODDE | LabX | Lucullus | USP | DSP | PAT | HPLC | GC | fermentation | cell culture | PCR

Team leadership | project ownership | Science Communication | analytical thinking